Training v2/v3 Report

Overview

This report documents the results of Phase 3 Version 3, which focuses on developing a balanced error-classification model using SMOTE oversampling. The goal of this training round was to combat class imbalance, reduce overfitting, and evaluate a more stable Random Forest configuration.

Dataset Summary

A total of **224 samples** were successfully loaded.

- **Feature Shape:** 224 × 28
- **Failed Files:** 0

Class Distribution (Before Balancing)

- good: 80 (35.7%)
- off_balance: 55 (24.6%)
- head_falling: 39 (17.4%)
- low_elbow: 27 (12.1%)
- late_foot: 23 (10.3%)

Imbalance Ratio: 3.48× — significant imbalance, justifying the use of SMOTE.

Train/Test Split

A stratified split was performed.

- **Training Samples:** 179
- **Testing Samples:** 45

Training Distribution

- good: 64
- head_falling: 31
- late_foot: 18

- low_elbow: 22
- off_balance: 44

Testing Distribution

- good: 16
- head_falling: 8
- late_foot: 5
- low_elbow: 5
- off_balance: 11

Feature Scaling

Features were standardized using **StandardScaler**.

Example values:

- Mean (first 5 features): 118.91, 121.72, 150.29, 151.13, 55.91
- Std (first 5 features): 25.08, 23.74, 21.00, 17.26, 30.03

SMOTE Oversampling

SMOTE (k_neighbors=5) was applied to the training set.

- **Before SMOTE:** 179 samples
- **After SMOTE:** 320 samples (64 per class)
- **New Samples Generated:** 141

All classes were perfectly balanced after oversampling.

Model Training

A Random Forest classifier was trained with anti-overfitting settings.

Configuration

- Trees: 150

- Max Depth: 12
- Min Samples Split: 10
- Min Samples Leaf: 5
- Class Weight: Balanced

Out-of-Bag Score: 65.00%

Training ran successfully.

Model Evaluation

Accuracy & F1 Scores

Metric	Training	Testing	Gap
Accuracy	96.09%	22.22%	73.87%
F1 (Weighted)	96.06%	22.50%	73.57%

A severe train-test gap confirms substantial **overfitting**. The model memorizes SMOTE-augmented training data but generalizes poorly.

Classification Report

- **good:** P=0.29, R=0.31, F1=0.30
- **head_falling:** P=0.29, R=0.25, F1=0.27
- **late_foot:** P=0.00, R=0.00, F1=0.00
- **low_elbow:** P=0.00, R=0.00, F1=0.00
- **off_balance:** P=0.30, R=0.27, F1=0.29

Per-Class Notes

- "late_foot" and "low_elbow" remain extremely difficult for the model.
- Overfitting obscures any benefit gained from balancing.

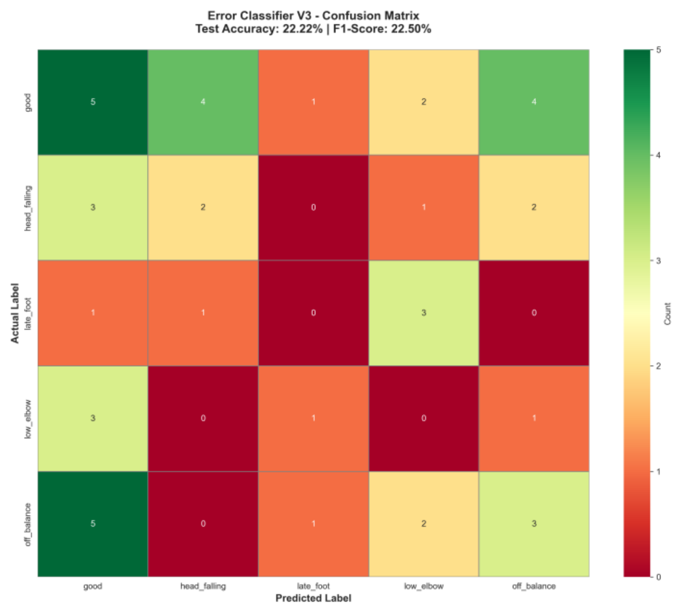
Cross-Validation

- **5-Fold CV Accuracy:** 64.38% (±7.42%)

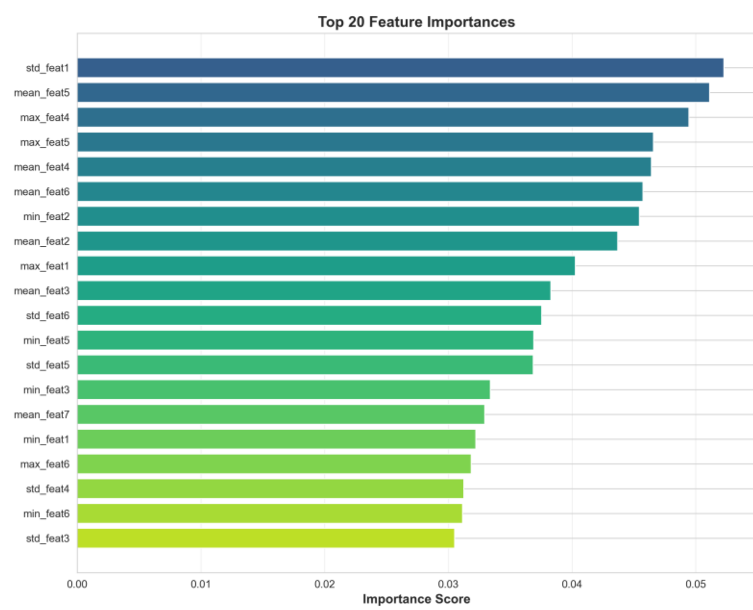
CV performance is far more stable, confirming the model is trainable but the test set behavior reveals generalization issues.

Visualizations Generated

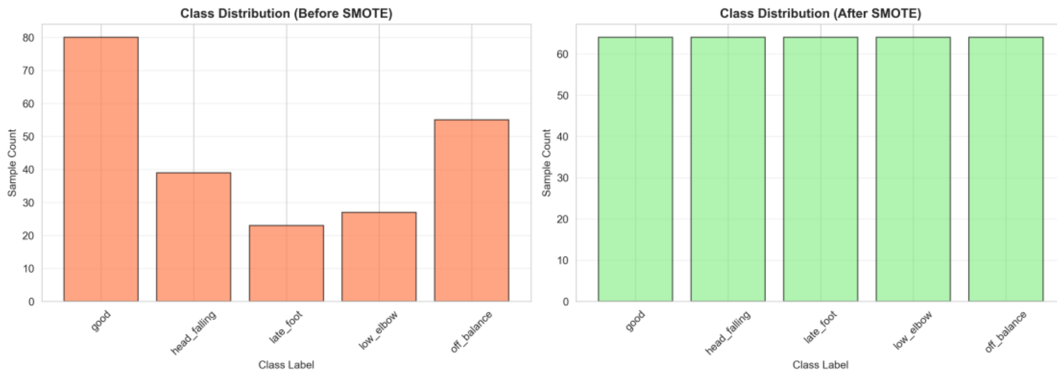
- **Confusion Matrix:**



- **Feature Importance Plot:**



- **Class Distribution Comparison Plot:**



Final Summary

Training v2/v3 produced a balanced dataset and improved internal consistency but still suffered from heavy overfitting.

Final Metrics

- **Test Accuracy:** 22.22%
- **Test F1 Score:** 22.50%
- **Overfitting:** 73.87% gap
- **Classes:** 5
- **SMOTE Applied:** Yes

Recommended Next Steps

1. Increase original dataset size.
2. Replace Random Forest with a simpler model (e.g., Logistic Regression) to test linear separability.
3. Experiment with PCA or feature selection.
4. Validate on new video samples.
5. Integrate with backend after refinement.

This concludes Training V2/V3 for Phase 3.